

Chenyi Xu

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Education

Hangzhou Dianzi University

B.Sc. in Computer Science and Technology GPA: 4.1 / 5.0

Core Courses: Advanced Mathematics, Probability Theory, Linear Algebra, Software Engineering, Computer Networks

Hangzhou, China

2023.09 – 2027.06

Research & Projects

4D Vision Reconstruction

Visiting Student, Collaborated with AGI Lab, Westlake University

Hangzhou, China

2025.07 – 2025.09

- Proposed a monocular dynamic video 4D reconstruction framework based on 4DGS, using single-step diffusion models and information-gain-driven pose selection to generate high-quality pseudo-views for multi-view supervision.
- Modeled scene dynamics using a canonical-space 4DGS representation and a globally shared $SE(3)$ motion basis trajectory, integrating a Neural Texture Field to achieve stable long-sequence reconstruction.

Publications

Yan, L.[†], Wu, Y., Xu, C., Yang, C. & Li, P. [AR-Nav Benchmark: Augmented Reality Navigation with Vision and Language. The 40th AAAI Conference on Artificial Intelligence \(AAAI 2026, Oral\)](#). ([†]Mentor).

- Proposed the AR-VLM model to construct high-fidelity 3D semantic fields via symbolic geometric alignment, resolving the long-term memory deficiency in AR navigation.
- Developed the ARN-Pilot module for lightweight topological reasoning and sub-second path planning, establishing the first standardized multi-scenario AR-Nav benchmark framework.

Yang, M., Yang, Y., Xu, C., Song, C., Zuo, Y., Zhao, T., Li, R., & Zhang, C. [Fast3Dcache: Training-free 3D Geometry Synthesis Acceleration. IEEE/CVF Conference on Computer Vision and Pattern Recognition \(CVPR 2026, Poster\)](#).

- Proposed Fast3Dcache, a training-free 3D geometric caching framework leveraging voxel stabilization regularities during 3D diffusion generation to accelerate inference.
- Designed dynamic cache scheduling (PCSC) and stability discrimination (SSC) based on motion features, achieving a 27% speedup and 54% FLOPs reduction while preserving geometric quality.

Xu, C., Wu, Y., Yan, L.[†], Yang, C., Zhang, J., Guan, F. & Li, P. LongDPM: Overlap-Aware 4D Reconstruction from Long Monocular Videos. (<https://arxiv.org/abs/2605.17303>)

- Proposed the LongDPM framework for long video 4D reconstruction via overlap-aware chunked inference, separating static anchors from dynamic features to ensure robust cross-chunk registration.
- Developed a geometrically-conditioned trajectory association and fusion mechanism to maintain identity continuity across chunks, achieving SOTA performance across multiple benchmarks.

Wu, Y.*^{*}, Xu, C.*^{*}, Yan, L., Cai, C., Min, G., Guan, F., Lin, B., Zhang, J. & Li, P. BrainNav: Towards Dual-Brain Minimal Sufficient Representation for Vision-Language Navigation. (*Equal contribution). (**NeurIPS 2026, Under Review**)

- Proposed the BrainNav dual-brain framework for long-horizon navigation, utilizing instruction-driven attention filtering to eliminate irrelevant visual noise and enhance language-spatial alignment.
- Constructed a low-rank constrained alignment and a compact world model for state prediction within compressed latent spaces, achieving SOTA performance on R2R and RxR benchmarks.

Wu, Y.*^{*}, Xu, C.*^{*}, Yan, L.[†], Bai, Z., Wang, Q., Guan, F., Zhang, J. & Li, P. Dense Soccer Video Captioning with Global Context and Player Position Awareness. (*Equal contribution). (**Scientific Reports, Under Review**)

- Proposed the SCPR framework for dense soccer video captioning, leveraging Perceiver attention (GCPR/PCPR modules) to fuse global-local video context with multi-modal player positions.
- Integrated an SAA spatial enhancement module to inject pitch tactical statistics for LLM decoding, yielding optimal fine-grained event localization and SOTA performance on public benchmarks.

Awards, Honors & Competitions

CUMCM “Higher Education Press Cup” | Provincial First Prize, National Third Prize

2024.09

Service Outsourcing Innovation & Entrepreneurship Competition | National Third Prize

2024.12 – 2025.03

ICT Industry-Education Integration Innovation Competition | National Second Prize

2024.08 – 2024.10

ICT Industry-Education Integration Innovation Competition | National Second Prize

2025.08 – 2025.10

Zhejiang Provincial Xinmiao Talent Program | Project Leader

2025.12 – 2026.12

First-Class Academic Scholarship & Outstanding Student Leader Award | HDU

2024 & 2025

Skills & Leadership

- Campus Leadership:** Class Monitor (Class 23052314) — Class Assistant (Class 25050417).
- Languages:** English (CET-4 & CET-6 qualified).
- Intellectual Property:** 5 Invention Patents filed/authorized.